**Project Title**: A Temporal Football Knowledge Graph for Tactical Pattern Discovery and Team Style Comparison

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**Mode**: ☒ 6 ECTS / ☐ 3 ECTS

**Motivation** Football matches are shaped by many interacting elements such as players, teams, events, possessions, field zones, and temporal sequences. Because tactical patterns and team identities emerge from these interactions over time, football data offers an interesting setting for a Knowledge Graph project.

**Problem**. Football event data is rich in temporal and relational structure, but it is typically analyzed in tabular form. While this works for basic statistics, it makes it difficult to explicitly model and compare recurring tactical patterns that depend on sequences of events or multiple interacting entities. The core problem of this project is therefore how to represent football match events as a temporal Knowledge Graph in a way that enables tactical pattern discovery and team style comparison.

**Solution.** Our idea is to build a temporal Knowledge Graph based on open football event data. The graph will include entities such as matches, teams, players, events, possessions, formations, and pitch zones, as well as relations that capture participation, spatial context, and temporal order. Based on this representation, we want to define and query a small set of tactical patterns, such as pressing regains, fast transitions, or build-up sequences in wide areas. This should allow us to use the graph not only to represent football match data in a structured way, but also to compare how different teams repeatedly realize such patterns.

**Steps: (Dataset: https://github.com/statsbomb/open-data)**

1. Get familiar with the Football Dataset and select suitable subset.
2. Create KG structure and create mappings from JSON data to KG representation.
3. Select suitable tactical pattern(s) and formulate corresponding queries

**Scope**. The dataset contains roughly 13 GB of JSON data with basic match information being small enough to process in full, but the event data is extensive. We will iteratively preprocess and map event types into the KG, starting with a core subset of the most important ones(e.g. passes, shots) and expanding the scope as time allows. The main effort goes into formulating tactical patterns as KG queries, again starting simple and increasing complexity as time allows.

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| Learning outcomes we plan to **focus** on | LO7, LO11 |
| Learning outcomes we plan to **show basic proficiency in** | LO1, LO2, LO4-6, LO8-9, LO12 |
| Learning outcomes we plan **not to include** | LO3, LO10 |